Linear-degree Sum-of-Squares refutation upper bound for false circuit-size claims

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doi:10.4230/LIPIcs.CCC.2023.31; CC BY4.0

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Editorial note. Selected complete native author main theorem/proof/context, original circuit diagrams and bibliography in standalone layout. Input files explicitly flattened into source spans, unused frontmatter/font packages omitted. No mathematical changes or new proofs. Original CC BY4.0 license retained.

Allowed standard prerequisites (editorial).

- Sum-of-Squares derivations, Boolean-cube polynomial identities and circuit polynomial encoding. Full author preliminaries and all Circuit axioms retained.

2 Preliminaries

All logarithms are in base 2. For integers $n \geq 1$ we write $[n] = \{1, 2, \ldots, n\}$ and for a set U we denote the power set of U by 2^U. Further, for a set $V \subseteq U$ we let $\overline{V}$ be the complement of V with respect to U, that is, $\overline{V} = U \setminus V$. We write $\binom{[n]}{\ell} \subseteq \{0,1\}^n$ for the set of binary strings with Hamming weight ℓ. For a string $\alpha \in \{0,1\}^n$ we let $|\alpha| = \sum_{i \in [n]} \alpha_i$.

We sometimes want to supress dependencies on constants and write $f(n, \varepsilon) \in O_\varepsilon\big(g(n, \varepsilon)\big)$, respectively $f(n, \varepsilon) \in \Omega_\varepsilon\big(g(n, \varepsilon)\big)$, to mean that there exists a function $c(\varepsilon) > 0$ such that there is an n_0 and for all $n \geq n_0$ it holds that $f(n, \varepsilon) \leq c(\varepsilon) \cdot g(n, \varepsilon)$, respectively $f(n, \varepsilon) \geq c(\varepsilon) \cdot g(n, \varepsilon)$.

Definition 2.1. A sequence of bipartite graphs $\{G_n = (U_n, V_n, E_n)\}_{n \in \mathbb{N}}$ with $\deg(u) = d$ for all $u \in U_n$ is *explicit* if there is an algorithm that given (n, u, j), where $n \in \mathbb{N}, u \in U_n$ and $j \in [d]$, computes the jth neighbor of vertex u in the graph G_n in time $\mathsf{poly}(\log n + \log |U| + \log d)$.

From now on it is understood that whenever we talk about an explicit graph we actually mean to say that there is a sequence of explicit graphs with above properties.

For a graph $G = (V, E)$ and $W \subseteq V$ we denote by

$$N(W) = \{\, u \in V \setminus W \mid (w, u) \in E \text{ for some } w \in W \,\}$$

the set of neighbors of W.

Definition 2.2. A bipartite graph $G = (U, V, E)$ is an (r, d, c)-expander if every vertex $u \in U$ has degree $\deg(u) = d$ and every set $W \subseteq U$ of size $|W| \leq r$ satisfies $|N(W)| \geq c \cdot |W|$.

A key ingredient in our proofs is the following result on the existence of strong explicit expanders.

Theorem 2.3 ([GUV09])**.** *For all constants $\gamma > 0$, every $M \in \mathbb{N}$, $r \leq M$, and $\varepsilon > 0$, there is an $N \leq d^2 \cdot r^{1+\gamma}$ and an explicit $(r, d, (1-\varepsilon)d)$-expander $G = (U, V, E)$, with $|U| = M$, $|V| = N$, and $d = O\big(((\log M)(\log r)/\varepsilon)^{1+1/\gamma}\big)$.*

For our purposes it is more relevant to compute the neighbor relation $\mathbf{Neigh}(u, v)$ indicating whether $(u, v) \in E$ rather than the neighbor function as in Definition 2.1, but this is an immediate consequence of being able to compute the neighbor function.

Claim 2.4. *If $G = (U, V, E)$ is explicit then the neighbor relation $\boldsymbol{Neigh} : U \times V \to \{0, 1\}$ is computable by a circuit of size $d \cdot \big(\mathsf{poly}(\log n + \log |U| + \log d) + 2 \log |V| + 1\big)$.*

A slice function is a Boolean function f such that there is a $\ell \in [n]$ with $f(\alpha) = 0$ whenever $|\alpha| < \ell$, and $f(\alpha) = 1$ whenever $|\alpha| > \ell$. Note that all slice functions are monotone.

The circuit complexity $\mathcal{C}(f)$ of a Boolean function f is the size of the smallest circuit over the basis $\vee, \wedge$, and $\neg$ (with fan-in 2). Similarly the monotone circuit complexity $\mathcal{C}_{\mathsf{mon}}(f)$ of a monotone Boolean function f is the size of the smallest circuit over the basis $\vee$, and $\wedge$. We have the following useful inequality between these measures.

Lemma 2.5 ([Ber82])**.** *If g is any slice function on n bits, then $\mathcal{C}_{\mathsf{mon}}(g) \leq 2\,\mathcal{C}(g) + O(n^2 \log n)$.*

Finally we also rely on the following simple claim.

Claim 2.6. *Let $p : \mathbb{R}^n \to \mathbb{R}$ be a degree d polynomial such that $p(x) = 0$ for all $x \in \{0, 1\}^n$. Then p can be written as*

$$p(x) = \sum_{i \in [n]} q_i(x) \cdot (x_i^2 - x_i)$$

where each term in the sum has degree at most d.

Proof sketch. We take the polynomial p and multilinearize it, using the appropriate polynomial $x_i^2 - x_i$. Eventually we are left with a sum of polynomials of the form $q_i(x) \cdot (x_i^2 - x_i)$ and a multilinear polynomial $\tilde{p}(x)$ which is 0 on all Boolean inputs. As multilinear polynomials are a basis for Boolean functions this implies that $\tilde{p}(x)$ is equal to the 0 polynomial and hence the claim follows. □

2.1 Sum of Squares

Let $\mathcal{P} = \{p_1 = 0, \ldots, p_m = 0\}$ be a system of polynomial equations over the set of variables $X = \{x_1, \ldots, x_n, \bar{x}_1, \ldots, \bar{x}_n\}$. Each p_i is called an *axiom*, and

throughout the paper we always assume that $\mathcal{P}$ includes all axioms $x_i^2 - x_i$ and $\bar{x}_i^2 - \bar{x}_i$, ensuring that the variables are Boolean, as well as the axioms $1 - x_i - \bar{x}_i$, making sure that the "bar" variables are in fact the negation of the "non-bar" variables.

Definition 2.7 (Sum-of-Squares)**.** *Sum-of-Squares (SoS)* is a static, semi-algebraic proof system. An SoS proof of $f \geq 0$ from $\mathcal{P}$ is a sequence of polynomials $\pi = (t_1, \ldots, t_m; s_1, \ldots, s_a)$ such that

$$\sum_{i \in [m]} t_i p_i + \sum_{i \in [a]} s_i^2 = f \ .$$

The *degree* of a proof π is $\mathsf{Deg}(\pi) = \max\{\max_{i \in [m]} \deg(t_i) + \deg(p_i), \max_{i \in [a]} 2\deg(s_i)\}$, an *SoS refutation of* $\mathcal{P}$ is an SoS proof of $-1 \geq 0$ from $\mathcal{P}$, and the *SoS degree to refute* $\mathcal{P}$ is the minimum degree of any SoS refutation of $\mathcal{P}$: if we let π range over all SoS refutations of $\mathcal{P}$, we can write $\mathsf{Deg}(\mathcal{P} \vdash_{\text{SoS}} \perp) = \min_\pi \mathsf{Deg}(\pi)$. The *size of an SoS refutation* π, $\mathsf{Size}(\pi)$, is the sum of the number of monomials in each polynomial in π and the *size of refuting* $\mathcal{P}$ is the minimum size over all refutations $\mathsf{Size}(\mathcal{P} \vdash_{\text{SoS}} \perp) = \min_\pi \mathsf{Size}(\pi)$.

Let us recall some well-known results about SoS. Given a bipartite graph $G = (U, V, E)$, and $b \in \{0,1\}^{|U|}$ we denote by $\Phi(G, b)$ the following XOR-CSP instance defined over G: for each $v \in V$ there is a Boolean variable x_v, and for every vertex $u \in U$ there is a constraint $\oplus_{v \in N(u)} x_v = b_u$. We encode this in the obvious way as a system of polynomial equations:

$$\Big\{ \prod_{v \in N(u)} (1 - 2 \cdot x_v) = 1 - 2 \cdot b_u \mid u \in U \Big\} \ ,$$

along with the Boolean axioms and the negation axioms for the x variables. The first theorem we need to recall is the classic lower bounds for XOR-CSPs by Grigoriev.

Theorem 2.8 ([Gri01])**.** *For $n \in \mathbb{N}$, all $k = k(n)$ and $r = r(n)$ the following holds. Let $G = (U, V, E)$ be an $(r, k, 2)$-expander with $|V| = n$. Then for every $b \in \{0,1\}^{|U|}$ SoS requires degree $\Omega(r)$ to refute the claim that there is a satisfying assignment to $\Phi(G, b)$.*

We also need to recall the size-degree tradeoff by Atserias and Hakoniemi.

Theorem 2.9 ([AH19])**.** *Let $\mathcal{P}$ be a system of polynomial equations over n Boolean variables and degree at most k. If d is the minimum degree SoS requires to refute $\mathcal{P}$, then the minimum size of an SoS refutation of $\mathcal{P}$ is at least $\exp(\Omega((d-k)^2/n))$.*

2.2 Restrictions

Let $\mathcal{P} = \{p_1 = 0, \ldots, p_m = 0\}$ be a system of polynomial equations over the set of Boolean variables $X = \{x_1, \ldots, x_n, \bar{x}_1, \ldots, \bar{x}_n\}$. For a map $\rho : \{x_1, \ldots, x_n\} \to \{0, 1, x_1, \ldots, x_n, \bar{x}_1, \ldots, \bar{x}_n\}$ denote by $\mathcal{P}\big|_\rho$ the system of polynomial equations

$\mathcal{P}$ restricted by ρ, i.e.,

$$\begin{aligned}\mathcal{P}\big|_\rho = \{&p_1(\rho(x_1),\dots,\rho(x_n)) = 0,\\ &p_2(\rho(x_1),\dots,\rho(x_n)) = 0,\\ &\vdots\\ &p_m(\rho(x_1),\dots,\rho(x_n)) = 0\}\ ,\end{aligned}$$

where it is understood that $\rho(\bar{x}_i) = \overline{\rho(x_i)}$, with the convention $\bar{\bar{x}}_i = x_i$, $\bar{0} = 1$ and vice versa. Throughout the paper all our restrictions set the bar variables to the negation of the non-bar variables. As such it makes sense to treat the pair of variables $(x_i, \bar{x}_i)$ as one variable and we say that $\mathcal{P}$ has n *unset* variables.

Definition 2.10 (Variable Substitution)**.** We say that a system of polynomial equations $\mathcal{P}'$ is a *variable substitution of* $\mathcal{P}$ if there is a map $\rho : \{x_1,\dots,x_n\} \to \{0,1,x_1,\dots,x_n,\bar{x}_1,\dots,\bar{x}_n\}$ such that $\mathcal{P}' = \mathcal{P}\big|_\rho$, where we ignore polynomial equations of the form $0 = 0$.

The following well-known lemma states that a system of polynomial equations $\mathcal{P}$ is at least as hard as any of its variable substitutions.

Lemma 2.11. *Let $\mathcal{P}, \mathcal{P}'$ be systems of polynomial equations such that $\mathcal{P}'$ is a variable substitution of $\mathcal{P}$. Then,*

(i) $\mathsf{Deg}(\mathcal{P} \vdash_{SoS} \bot) \geq \mathsf{Deg}(\mathcal{P}' \vdash_{SoS} \bot)$*, and*

(ii) $\mathsf{Size}(\mathcal{P} \vdash_{SoS} \bot) \geq \mathsf{Size}(\mathcal{P}' \vdash_{SoS} \bot)$.

The lemma is easy to verify by considering an SoS refutation of $\mathcal{P}$ and hitting it with the appropriate variable substitution. The restricted proof is now a refutation of $\mathcal{P}'$ and it can be seen that the degree/size of the restricted refutation is at most the degree/size of the original refutation.

We also consider more general substitutions.

Definition 2.12 (Polynomial Substitution)**.** Functions $\rho : \{x_1,\dots,x_n\} \to \mathbb{R}[x]_{\leq k}$ that map variables to polynomials of degree at most k are called *polynomial substitutions*.

For polynomial substitutions we have the following well-known lemma.

Lemma 2.13. *Let $\mathcal{P}$ be a system of polynomial equations and let ρ be a polynomial substitution mapping variables to polynomials of degree at most k. Then,* $\mathsf{Deg}(\mathcal{P} \vdash_{SoS} \bot) \geq \mathsf{Deg}(\mathcal{P}\big|_\rho \vdash_{SoS} \bot)/k$.

This lemma can again be verified by considering a refutation of $\mathcal{P}$. Substitute each variable x_i in the proof by $\rho(x_i)$. This results in a refutation of $\mathcal{P}\big|_\rho$, whose degree is at most a factor k larger than the degree of the refutation of $\mathcal{P}$.

2.3 The Circuit Size Formula

The formula $\mathrm{Circuit}_s(f)$ encodes the claim that the function f, given as a truthtable $f \in \{0,1\}^{2^n}$, can be computed by a circuit of size s over n Boolean inputs $x_1,\dots,x_n$. The encoding is not essential but for concreteness let us fix

one encoding of this claim. We deviate from the encoding used by Razborov [Raz98, Raz04b] and do not present the formula as a propositional formula but rather as a system of polynomial equations. In order to encode below constraints as a constant width CNF formula, as done by Razborov, one needs to introduce extension variables. Despite this difference it is not difficult to see that our lower bound also works against the CNF encoding. In Section 3 we directly show that a low degree SoS refutation of the CNF encoding gives rise to a low degree SoS refutation of the encoding used in this paper (see Proposition 3.1). Thus a lower bound for our encoding implies a lower bound for the CNF encoding. As the presentation is simpler in the polynomial encoding, we present it as follows.

We also need to define the monotone version of $\mathrm{Circuit}_s(f)$ denoted by $\mathrm{Circuit}_s^{\mathsf{mon}}(f)$. The later is a restriction of the former with the $\mathsf{IsNeg}(v)$ (see below) variable, for all $v \in [s]$, set to 0. This forces the circuit to only contain $\wedge$ and $\vee$ gates, i.e., the circuit is monotone.

All variables introduced in the following are Boolean variables and we implicitly add the Boolean axiom $y(1-y) = 0$ for each variable y and further implicitly introduce the "bar variable" $\bar{y}$ along with the negation axiom $y = 1 - \bar{y}$ (and the corresponding Boolean axiom) ensuring that $\bar{y}$ is always the negation of y.

Let us first describe the *structure variables* which are used to describe the circuit that supposedly computes the function f.

We view the s gates as being indexed from 1 to s in topological order with gate s being the output. For each gate $v \in [s]$ there are three variables $\mathsf{IsNeg}(v), \mathsf{IsOr}(v), \mathsf{IsAnd}(v)$ indicating the operation computed at v. Similarly for a gate $v \in [s]$ and a wire $a \in \{1,2\}$ we have variables $\mathsf{IsFromConst}(v,a), \mathsf{IsFromInput}(v,a), \mathsf{IsFromGate}(v,a)$ indicating whether the input wire a of v is connected to a constant, a variable or a gate.

Further, we have the variables $\mathsf{ConstantValue}(v,a)$, $\mathsf{IsInput}(v,a,i)$ and $\mathsf{IsGate}(v,a,u)$, for $a \in \{1,2\}$, $i \in [n]$ and $u < v$, specifying the constant value, input x_i or gate u, the input wire a of v is connected to (assuming a is connected to the corresponding kind).

The second set of variables are the *evaluation variables*, which describe what value is computed at each v on input $\alpha = \alpha_1, \ldots, \alpha_n$ (i.e., we have $x_i = \alpha_i$).

For each gate $v \in [s]$ and assignment $\alpha \in \{0,1\}^n$ we have a Boolean variable $\mathsf{Out}_\alpha(v)$ indicating the value computed at gate v on input α. The Boolean variable $\mathsf{In}_\alpha(v,a)$ indicates the value brought to the vertex $v \in [s]$ on wire $a \in \{1,2\}$ on input α.

Note that there is a total of $3s + 6s + 2s + 2sn + 2\binom{s}{2} = \Theta(s^2 + sn)$ structure variables, and a total of $3s2^n$ evaluation variables, for a total of $\Theta(s^2 + s2^n)$ variables in $\mathrm{Circuit}_s(f)$.

The formula consists of the following axioms. For the sake of readability we omit some universal quantifiers: the variable $a \in \{1,2\}$ in Equations (1) and (3) to (9) as well as the variable $\alpha \in \{0,1\}^n$ in Equations (7) to (13) are implicitly universally quantified.

Let us first describe the axioms on the structure of the circuit. In the following section we refer to this set of axioms as the *structure axioms*. The first axioms ensure that every wire is connected to a single kind

$$\mathsf{IsFromConst}(v,a) + \mathsf{IsFromInput}(v,a) + \mathsf{IsFromGate}(v,a) = 1 \quad \forall\, v \in [s] \ , \quad (1)$$

and similarly the next axioms make sure that each gate is of precisely one kind

$$\mathsf{IsNeg}(v) + \mathsf{IsOr}(v) + \mathsf{IsAnd}(v) = 1 \quad \forall\, v \in [s] \ . \tag{2}$$

The final structure axioms ensure that the variables, which indicate to what input or gate a fixed wire is connected to, always sum to one (except for gate 1 which cannot have any inputs from other gates)

$$\sum_{i=1}^{n} \mathsf{IsInput}(v,a,i) = 1 \quad \forall v \in [s], \text{ and} \tag{3}$$

$$\sum_{u=1}^{v-1} \mathsf{IsGate}(v,a,u) = 1 \quad \forall v \in [s] \setminus \{1\} \ . \tag{4}$$

We further strengthen our encoding by adding the axioms

$$\mathsf{IsInput}(v,a,i)\,\mathsf{IsInput}(v,a,j) = 0 \quad \forall v \in [s],\ i < j \in [n], \text{ and} \tag{5}$$

$$\mathsf{IsGate}(v,a,u)\,\mathsf{IsGate}(v,a,u') = 0 \quad \forall u < u' < v \in [s] \ . \tag{6}$$

Note that Equations (5) and (6) are implied by Equations (3) and (4). We add these axioms in order to argue that a short refutation of the CNF encoding of this principle leads to a short refutation of the present encoding.

The second group of axioms are the *evaluation axioms* and they ensure that the evaluation variables indeed compute the intended values. We start by making sure that the wires carry the value intended by the structure axioms. If a wire is connected to a constant, then the evaluation variable associated with that wire should always be equal to the constant

$$\mathsf{IsFromConst}(v,a) \cdot \big(\mathsf{In}_\alpha(v,a) - \mathsf{ConstantValue}(v,a)\big) = 0 \ , \tag{7}$$

and similarly in case if a wire is connected to an input or a gate

$$\mathsf{IsFromInput}(v,a) \cdot \mathsf{IsInput}(v,a,i) \cdot \big(\mathsf{In}_\alpha(v,a) - \alpha_i\big) = 0 \ , \tag{8}$$

$$\mathsf{IsFromGate}(v,a) \cdot \mathsf{IsGate}(v,a,u) \cdot \big(\mathsf{In}_\alpha(v,a) - \mathsf{Out}_\alpha(u)\big) = 0 \ . \tag{9}$$

The final set of evaluation axioms makes sure that the output evaluation variable of a gate is correctly related to the input evaluation variables:

$$\mathsf{IsNeg}(v) \cdot \mathsf{Out}_\alpha(v) = \mathsf{IsNeg}(v) \cdot \overline{\mathsf{In}_\alpha(v,1)} \ , \tag{10}$$

$$\mathsf{IsOr}(v) \cdot \mathsf{Out}_\alpha(v) = \mathsf{IsOr}(v) \cdot \big(1 - \overline{\mathsf{In}_\alpha(v,1)} \cdot \overline{\mathsf{In}_\alpha(v,2)}\big) \ , \tag{11}$$

$$\mathsf{IsAnd}(v) \cdot \mathsf{Out}_\alpha(v) = \mathsf{IsAnd}(v) \cdot \mathsf{In}_\alpha(v,1) \cdot \mathsf{In}_\alpha(v,2) \ . \tag{12}$$

Last but not least we have the axioms that ensure that the circuit outputs the function specified by the truthtable

$$\mathsf{Out}_\alpha(s) = f(\alpha) \ . \tag{13}$$

Theorem 2.14. *For all $\varepsilon > 0$ there is a $d = d(\varepsilon)$ such that the following holds. For $n \in \mathbb{N}$, all $s \geq n^d$ and any Boolean function $f : \{0,1\}^n \to \{0,1\}$ on n bits, SoS requires degree $\Omega_\varepsilon(s^{1-\varepsilon})$ to refute* $\mathrm{Circuit}_s(f)$.

It is worthwhile to point out that the proof of Theorem 2.14 is not specific to the SoS proof system. In fact we outline a general reduction that shows that if one has a CSP lower bound of the form of Theorem 2.8 that only requires good expansion of the underlying constraint-variable incidence graph and the proof system is expressive enough so that one can replace variables by local Boolean functions, then one obtains strong lower bounds for the $\text{Circuit}_s(f)$ formula.

The lower bound of $\Omega_\varepsilon(s^{1-\varepsilon})$ on the degree is essentially tight: if f does not have a circuit of size s then there exists an SoS refutation of this statement in degree $O(s)$.

3 On Encodings of the $\text{Circuit}_s(f)$ Tautology

Let us introduce a possible constant width CNF encoding of $\text{Circuit}_s(f)$ as proposed by Razborov [Raz98].

The formula is defined over the same variables as introduced in Section 2.3, but in order to keep the fan-in bounded, we further introduce the extension variables $\mathsf{IsInput}^{\leq}(v,a,i)$ and $\mathsf{IsGate}^{\leq}(v,a,u)$ that indicate whether the wire a of v is connected to a variable in $x_1,\ldots,x_i$, a gate $1,\ldots,u$ respectively.

Let us group the axioms in the same manner as we did in Section 2.3. First we have the structure axioms. The first axioms encode that each wire is connected to a single kind

$$\begin{aligned}
&\big(\mathsf{IsFromConst}(v,a) \vee \mathsf{IsFromInput}(v,a) \vee \mathsf{IsFromGate}(v,a)\big)\wedge\\
&\neg\big(\mathsf{IsFromConst}(v,a) \wedge \mathsf{IsFromInput}(v,a)\big)\wedge\\
&\neg\big(\mathsf{IsFromInput}(v,a) \wedge \mathsf{IsFromGate}(v,a)\big)\wedge\\
&\neg\big(\mathsf{IsFromConst}(v,a) \wedge \mathsf{IsFromGate}(v,a)\big)\ . \qquad (14)
\end{aligned}$$

The next set of axioms similarly ensures that each gate computes precisely one function

$$\begin{aligned}
&\big(\mathsf{IsNeg}(v) \vee \mathsf{IsOr}(v) \vee \mathsf{IsAnd}(v)\big)\wedge\\
&\neg\big(\mathsf{IsNeg}(v) \wedge \mathsf{IsOr}(v)\big) \wedge \neg\big(\mathsf{IsOr}(v) \wedge \mathsf{IsAnd}(v)\big) \wedge \neg\big(\mathsf{IsNeg}(v) \wedge \mathsf{IsAnd}(v)\big)\ . \qquad (15)
\end{aligned}$$

Last, we need to make sure that each wire is connected to a single input or a gate.

$$\begin{aligned}
&\mathsf{IsInput}^{\leq}(v,a,n) \wedge \bigwedge_{i\neq j} \neg\big(\mathsf{IsInput}(v,a,i) \wedge \mathsf{IsInput}(v,a,j)\big)\wedge\\
&\bigwedge_{i\in[n]} \Big(\mathsf{IsInput}^{\leq}(v,a,i) \equiv \big(\mathsf{IsInput}^{\leq}(v,a,i-1) \vee \mathsf{IsInput}(v,a,i)\big)\Big)\ ,\\
&\qquad \text{where } \mathsf{IsInput}^{\leq}(v,a,0) \stackrel{\text{def}}{=} 0\ , \qquad (16)
\end{aligned}$$

and similarly for $v \in [s] \setminus \{1\}$ we have that

$$\mathsf{IsGate}^{\leq}(v,a,v-1) \wedge \bigwedge_{u<u'<v} \neg\big(\mathsf{IsGate}(v,a,u) \wedge \mathsf{IsGate}(v,a,u')\big) \wedge$$
$$\bigwedge_{u\in[v-1]} \Big(\mathsf{IsGate}^{\leq}(v,a,u) \equiv \big(\mathsf{IsGate}^{\leq}(v,a,u-1) \vee \mathsf{IsGate}(v,a,u)\big)\Big) \ ,$$
$$\text{where } \mathsf{IsGate}^{\leq}(v,a,0) \stackrel{\text{def}}{=} 0 \ . \tag{17}$$

Let us take a look at the evaluation axioms. Again, we have axioms that ensure that the wires carry the values intended by the structure variables. If a wire is connected to a constant, then the evaluation variable associated with that wire should be equal to the constant

$$\mathsf{IsFromConst}(v,a) \to \big(\mathsf{In}_\alpha(v,a) \equiv \mathsf{ConstantValue}(v,a)\big) \ , \tag{18}$$

and similarly if a wire is connected to an input or a gate

$$\mathsf{IsFromInput}(v,a) \wedge \mathsf{IsInput}(v,a,i) \to \mathsf{In}_\alpha(v,a) \equiv \alpha_i \ , \tag{19}$$
$$\mathsf{IsFromGate}(v,a) \wedge \mathsf{IsGate}(v,a,u) \to \mathsf{In}_\alpha(v,a) \equiv \mathsf{Out}_\alpha(u) \ . \tag{20}$$

Last we need to make sure that the gates propagate the value they are supposed to compute.

$$\mathsf{IsNeg}(v) \to \big(\mathsf{Out}_\alpha(v) \equiv \neg\,\mathsf{In}_\alpha(v,1)\big) \tag{21}$$
$$\mathsf{IsOr}(v) \to \big(\mathsf{Out}_\alpha(v) \equiv \mathsf{In}_\alpha(v,1) \vee \mathsf{In}_\alpha(v,2)\big) \tag{22}$$
$$\mathsf{IsAnd}(v) \to \big(\mathsf{Out}_\alpha(v) \equiv \mathsf{In}_\alpha(v,1) \wedge \mathsf{In}_\alpha(v,2)\big) \ . \tag{23}$$

The final axioms ensure that the correct function is computed

$$\mathsf{Out}_\alpha(s) \equiv f(\alpha) \ . \tag{24}$$

This formula can be rewritten in the usual manner into a 4-CNF. Let us denote this formula by $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)$.

Proposition 3.1. *If there is an SoS refutation of degree d of the CNF formula $\mathrm{Circuit}_s^{CNF}(f)$, then there is an SoS refutation of degree $O(d)$ of the system of polynomials $\mathrm{Circuit}_s(f)$ as introduced in Section 2.3.*

The rest of this section is devoted to the proof of Proposition 3.1.

Observe that for each axiom p from the polynomial encoding $\mathrm{Circuit}_s(f)$, there is a CNF $F_p \subseteq \mathrm{Circuit}_s^{\mathrm{CNF}}(f)$ over the same variables as p (ignoring the added extension variables $\mathsf{IsInput}^{\leq}(v,a,i)$ and $\mathsf{IsGate}^{\leq}(v,a,u)$) such that $p(\alpha) = 0$ is satisfied by a Boolean assignment α if and only if F_p is satisfied by α (where we extend the assignment to the extension variables in the natural manner).

Recall that SoS operates on polynomials and we thus need to translate the CNF into a system of polynomials. We translate a clause $\vee_{i\in[w]} z_i$ into the polynomial $\prod_{i\in[w]}(1-z_i) = 0$.

Observe that almost all axioms p of $\mathrm{Circuit}_s(f)$ depend only on a constant number of variables. From such p, using the appropriate Boolean axioms and negation axioms, we can in constant degree derive F_p.

Let us define a polynomial substitution ρ that gets rid of the extension variables in the natural manner: the substitution ρ first replace each occurrence of a "bar" extension variable $\overline{\mathsf{IsInput}^{\leq}(v,a,i)}$ or $\overline{\mathsf{IsGate}^{\leq}(v,a,u)}$ by the polynomial $1-\mathsf{IsInput}^{\leq}(v,a,i)$ and the polynomial $1-\mathsf{IsGate}^{\leq}(v,a,u)$ respectively. Then, ρ replaces each occurrence of the variable $\mathsf{IsInput}^{\leq}(v,a,i)$ by $\sum_{j\leq i}\mathsf{IsInput}(v,a,j)$ and similarly $\mathsf{IsGate}^{\leq}(v,a,u)$ by $\sum_{w\leq u}\mathsf{IsGate}(v,a,w)$.

Suppose we have a degree d refutation π of $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)$. Let us consider $\pi\big|_\rho$ and $\mathrm{Circuit}^{\mathrm{CNF}}(f)\big|_\rho$. Note that $\pi\big|_\rho$ is a degree d SoS refutation of $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)\big|_\rho$.

We claim that in constant degree the axioms of $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)\big|_\rho$ can be derived from the polynomial encoding $\mathrm{Circuit}_s(f)$. As previously noted, this holds for all axioms but the ones that are over a non-constant number of variables. In other words it just remains to show that we can derive the substituted Equations (16) and (17) from Equations (3) to (6).

Let us consider Equation (16). With the extension variables substituted and translated into a system of polynomials the axiom consists of the following polynomial equations.

$$1-\sum_{j\in[n]}\mathsf{IsInput}(v,a,j)=0 \tag{25}$$

$$\mathsf{IsInput}(v,a,i)\cdot\mathsf{IsInput}(v,a,j)=0,\ \text{for } i\neq j \tag{26}$$

$$\left(\sum_{j\leq i}\mathsf{IsInput}(v,a,j)\right)\left(1-\sum_{j<i}\mathsf{IsInput}(v,a,j)\right)\cdot \overline{\mathsf{IsInput}(v,a,i)}=0,\ \text{for } i\in[n] \tag{27}$$

$$\left(1-\sum_{j\leq i}\mathsf{IsInput}(v,a,j)\right)\left(\sum_{j<i}\mathsf{IsInput}(v,a,j)\right)=0,\ \text{for } i\in[n] \tag{28}$$

$$\left(1-\sum_{j\leq i}\mathsf{IsInput}(v,a,j)\right)\mathsf{IsInput}(v,a,i)=0,\ \text{for } i\in[n]\ . \tag{29}$$

Equation (25) is equal to Equation (3) and similarly Equation (26) is equal to Equation (5). In the following we show that Equations (27) to (29) can be derived from Equation (5), the Boolean axioms and the negation axioms in constant degree.

Consider Equation (27). Expand and rewrite modulo the Boolean axioms and the negation axiom to obtain

$$\mathsf{IsInput}(v,a,i)\left(\sum_{j<i}\mathsf{IsInput}(v,a,j)\right)^2 - 2\sum_{j<j'<i}\mathsf{IsInput}(v,a,j)\cdot\mathsf{IsInput}(v,a,j') - \mathsf{IsInput}(v,a,i)\sum_{j<i}\mathsf{IsInput}(v,a,j)=0\ . \tag{30}$$

Observe that every term t left in this polynomial is of the form $t = t' \cdot \mathsf{IsInput}(v, a, j) \cdot \mathsf{IsInput}(v, a, j')$, for some $j \neq j' \in [i]$ and a term t' of degree at most 1. But this means that every term is equal to 0 modulo Equation (5) and we thus see that Equation (27) can be derived in constant degree from $\text{Circuit}_s(f)$.

Let us consider Equation (28). Rewrite modulo the Boolean axiom to obtain

$$\mathsf{IsInput}(v, a, i) \sum_{j<i} \mathsf{IsInput}(v, a, j) + 2 \sum_{j<j'<i} \mathsf{IsInput}(v, a, j) \cdot \mathsf{IsInput}(v, a, j') = 0 \ . \tag{31}$$

All terms are of the form of Equation (5) and we can thus derive Equation (28) from $\text{Circuit}_s(f)$ in constant degree.

Last, we need to consider Equation (29). Note that modulo the Boolean axiom we obtain the polynomial equation

$$-\mathsf{IsInput}(v, a, i) \sum_{j<i} \mathsf{IsInput}(v, a, j) = 0 \ . \tag{32}$$

Also in this polynomial every term is of the form of Equation (5) and thus also Equation (29) can be derived in constant degree.

What remains is to show that Equation (17) can be derived from $\text{Circuit}_s(f)$ in constant degree. This can be checked analogous to Equation (16) and we thus omit it here.

We conclude that all axioms of $\text{Circuit}_s^{\text{CNF}}(f)\big|_\rho$ can be derived from $\text{Circuit}_s(f)$ in constant degree and thus a degree d SoS refutation of $\text{Circuit}_s^{\text{CNF}}(f)$ gives rise to a degree $O(d)$ SoS refutation of $\text{Circuit}_s(f)$. Equivalently, a degree d lower bound for $\text{Circuit}_s(f)$ implies a degree $\Omega(d)$ lower bound for $\text{Circuit}_s^{\text{CNF}}(f)$ as claimed.

Proposition 3.2. *Let $s \in N$ and $f : \{0,1\}^n \to \{0,1\}$ be a Boolean function on n bits that requires circuits of size larger than s to be computed. Then there is a degree $O(s)$ SoS refutation of* $\text{Circuit}_s(f)$.

4 Degree Upper Bound

In this section we give a simple upper bound on the SoS refutation degree for $\text{Circuit}_s(f)$. Specifically, for functions f that have no circuit of size s, we show that there is an SoS refutation of $\text{Circuit}_s(f)$ of degree $O(s)$, essentially matching our $\Omega(s^{1-\epsilon})$ lower bound.

At a high level, the logic behind the refutation is as follows: first, we show that SoS in degree $O(s)$ can derive that the $\Theta(s^2)$ structure variables of $\text{Circuit}_s(f)$ must uniquely describe a circuit of size s. Then, we also show that for any fixed circuit, SoS can in degree $O(s)$ derive that the circuit does not compute f by "evaluating" the circuit on some (non-deterministically chosen) input where the output differs from f.

To make this precise, we first define a set of monomials, which correspond to circuits of size s. A multilinear monomial m is a *circuit monomial* if for every gate $v \in [s]$ it holds that

1. exactly one of the variables $\mathsf{IsNeg}(v), \mathsf{IsOr}(v)$ or $\mathsf{IsAnd}(v)$ occurs in m,
2. for $a \in \{1,2\}$ exactly one of the variables $\mathsf{IsFromConst}(v,a)$, $\mathsf{IsFromInput}(v,a)$ or $\mathsf{IsFromGate}(v,a)$ occurs in m,
3. for $a \in \{1,2\}$ exactly one of the variables $\mathsf{ConstantValue}(v,a)$ or $\overline{\mathsf{ConstantValue}(v,a)}$ occurs in m,
4. for $a \in \{1,2\}$ exactly one of the variables $\{\mathsf{IsInput}(v,a,i) \mid i \in [n]\}$ occurs in m, and
5. for $a \in \{1,2\}$ and $v > 1$, exactly one of the variables $\{\mathsf{IsGate}(v,a,u) \mid u < v\}$ occurs in m, and
6. no other variables occur in m than the ones described above.

We denote by $\mathcal{M}_s$ the set of circuit monomials. We first show that SoS can derive in degree $O(s)$ the polynomial $\sum_{m\in\mathcal{M}_s} m - 1$; this corresponds to SoS proving that the structure variables uniquely describe a circuit and does not use anything about f. Then in a second step we show that for every $m \in \mathcal{M}_s$, SoS can derive $-m$ in degree $O(s)$; this corresponds to SoS proving that the circuit described by m does not compute f correctly. Summing these two parts up yields an SoS derivation of -1, i.e., a refutation of the $\mathrm{Circuit}_s(f)$ formula.

Deriving $\sum_{m\in\mathcal{M}_s} m - 1$. We proceed by induction on s. Note that $\mathcal{M}_0 = \{1\}$ and hence the base case is trivial. Suppose we have an SoS derivation of $\sum_{m\in\mathcal{M}_s} m - 1$. For every monomial $m \in \mathcal{M}_s$ we add the polynomial

$$m \cdot \big(\mathsf{IsNeg}(v) + \mathsf{IsOr}(v) + \mathsf{IsAnd}(v) - 1\big) \tag{33}$$

to the derivation (note that the second term is Equation (2)). This gives us an SoS derivation of $\sum_{m\in\mathcal{M}'_s} m - 1$, where

$$\mathcal{M}'_s = \bigcup_{m\in\mathcal{M}_s} \{m\cdot\mathsf{IsNeg}(v), m\cdot\mathsf{IsOr}(v), m\cdot\mathsf{IsAnd}(v)\} \ .$$

We can continue in the same manner with Equation (1) and Equation (3) to finally obtain an SoS derivation of $\sum_{m\in\mathcal{M}_{s+1}} m - 1$. Clearly this derivation requires degree at most $O(s)$, as for each gate there are at most 7 variables in every monomial from $\mathcal{M}_s$.

Deriving $-m$ for $m \in \mathcal{M}_s$. Let C be the circuit that corresponds to the monomial m and let $\alpha \in \{0,1\}^n$ be such that $f(\alpha) \neq C(\alpha)$ (by the assumption that f does not have a circuit of size s, such an α exists). Suppose $C(\alpha) = b$ but $f(\alpha) = 1 - b$.

We construct a degree $O(s)$ SoS proof of the fact that $C(\alpha) = b$. That is, we are going to show that the polynomial $p_s = m \cdot (\mathsf{Out}_\alpha(s) - b)$ can be written as

$$p_s = \sum_{i=1}^{t} r_i \cdot q_i \ , \tag{34}$$

for some parameter t, axioms $q_1, \ldots, q_t$ and some polynomials $r_1, \ldots, r_t$, such that $\deg(r_i \cdot q_i) = O(s)$ for all i. Note that given this, we can then easily derive $-m$ by subtracting the polynomial $m \cdot (\mathsf{Out}_\alpha(s) - f(\alpha)) = m \cdot (\mathsf{Out}_\alpha(s) - 1 + b)$ (this is m multiplied by Equation (13)), yielding a derivation of $m \cdot (1 - 2b)$ which is either m or $-m$ depending on b; in the former case it can be multiplied by -1 to yield $-m$. Note that here it is important that the derivation (34) is a Nullstellensatz derivation, not using any Sum-of-Squares part, since otherwise it would not be possible to multiply it by -1.

Let us thus see how to derive $m \cdot (\mathsf{Out}_\alpha(s) - b)$. We do this by structural induction over the circuit: for every gate v we are going to construct an SoS proof of the fact that the circuit rooted at v outputs the bit b_v on input α. In other words, an SoS derivation of the polynomial $m \cdot (\mathsf{Out}_\alpha(v) - b_v)$.

Let us explain how to construct an SoS proof p_v. Consider a gate v in the circuit. Depending on the function computed at v and how the wires of v are connected we construct p_v slightly differently. As a first step let us construct SoS proofs q_1 and q_2 of the fact that on input α the bit c_a, $a \in \{1, 2\}$, is carried on wire a to the gate v. That is, the polynomial q_a should simplify to $m \cdot \big(\mathsf{In}_\alpha(v, a) - c_a\big)$. In the following we explain how to precisely define q_a depending on what the wire is connected to. Note that not a lot is going on – we are mostly just multilinearizing using the Boolean axioms.

If m is of the form $m = m' \cdot \mathsf{IsFromConst}(v, a) \cdot \mathsf{ConstantValue}(v, a)$ for some monomial m', i.e., wire a is connected to the constant $c_a = 1$ in the circuit described by m, then we can derive $q_a = m \cdot \big(\mathsf{In}_\alpha(v, a) - 1\big)$ by the identity

$$\begin{aligned} q_a = m' \cdot \mathsf{ConstantValue}(v,a) \cdot \mathsf{IsFromConst}(v,a) \cdot \big(\mathsf{In}_\alpha(v,a) - \mathsf{ConstantValue}(v,a)\big) + \\ m' \cdot \mathsf{IsFromConst}(v,a) \cdot \big(\mathsf{ConstantValue}(v,a)^2 - \mathsf{ConstantValue}(v,a)\big) \ , \end{aligned} \tag{35}$$

a linear combination of Equation (7) and the Boolean axiom on $\mathsf{ConstantValue}(v, a)$. Similarly if $m = m' \cdot \mathsf{IsFromConst}(v, a) \cdot \overline{\mathsf{ConstantValue}(v, a)}$ (i.e., $c_a = 0$) we can derive $q_a = m \cdot \mathsf{In}_\alpha(v, a)$ by

$$\begin{aligned} q_a = m' \cdot \overline{\mathsf{ConstantValue}(v,a)} \cdot \mathsf{IsFromConst}(v,a) \cdot \big(\mathsf{In}_\alpha(v,a) - \mathsf{ConstantValue}(v,a)\big) + \\ m' \cdot \mathsf{IsFromConst}(v,a) \cdot \mathsf{ConstantValue}(v,a) \cdot \\ \big(1 - \mathsf{ConstantValue}(v,a) - \overline{\mathsf{ConstantValue}(v,a)}\big) + \\ m' \cdot \mathsf{IsFromConst}(v,a) \cdot (\mathsf{ConstantValue}(v,a)^2 - \mathsf{ConstantValue}(v,a)) \ , \end{aligned} \tag{36}$$

where we additionally use the negation axiom on $\mathsf{ConstantValue}(v, a)$.

Next, if a is connected to an input i and m is of the form $m = m' \cdot \mathsf{IsFromInput}(v, a) \cdot \mathsf{IsInput}(v, a, i)$ (so that $c_a = \alpha_i$), then

$$q_a = m \cdot (\mathsf{In}_\alpha(v,a) - \alpha_i) = m' \cdot \mathsf{IsFromInput}(v,a) \cdot \mathsf{IsInput}(v,a,i) \cdot \big(\mathsf{In}_\alpha(v,a) - \alpha_i\big) \ , \tag{37}$$

which is a multiple of Equation (8). Lastly, if a is connected to a gate u and m is of the form $m = m' \cdot \mathsf{IsFromGate}(v, a) \cdot \mathsf{IsGate}(v, a, u)$ (i.e., $c_a = b_u$), then

$$\begin{aligned} q_a = m \cdot (\mathsf{In}_\alpha(v,a) - b_u) = m' \cdot \mathsf{IsFromGate}(v,a) \cdot \mathsf{IsGate}(v,a,u) \cdot \big(\mathsf{In}_\alpha(v,a) - \mathsf{Out}_\alpha(u)\big) \\ + m \cdot (\mathsf{Out}_\alpha(u) - b_u) \ , \end{aligned} \tag{38}$$

where the first term is a multiple of Equation (9), and the second term is the polynomial p_u which, by induction, we assume has already been derived in degree $O(s)$.

Given the two SoS proofs q_1 and q_2 we are ready to construct the SoS proof $p_v = m \cdot (\mathsf{Out}_\alpha(v) - b_v)$. As mentioned earlier we do a case distinction over the funtion computed at v.

1. *v is a* not *gate ($m = m' \cdot \mathsf{IsNeg}(v)$ and $b_v = 1 - c_1$).* We have the derivation

$$\begin{aligned} p_v = m' \cdot \mathsf{IsNeg}(v) \cdot \big(\mathsf{Out}_\alpha(v) - \overline{\mathsf{In}_\alpha(v,1)} \big) \\ + m \cdot \big(\overline{\mathsf{In}_\alpha(v,1)} - 1 + \mathsf{In}_\alpha(v,1) \big) \\ - q_1 \ , \end{aligned} \tag{39}$$

   where the first line uses Equation (10) and the second line uses the negation axiom for $\mathsf{In}_\alpha(v,1)$.

2. *v is an* or *gate ($m = m' \cdot \mathsf{IsOr}(v)$, and $b_v = 1 - (1-c_1)(1-c_2)$.* We have the derivation

$$\begin{aligned} p_v = & m' \cdot \mathsf{IsOr}(v) \cdot \Big(\mathsf{Out}_\alpha(v) - \big(1 - \overline{\mathsf{In}_\alpha(v,1)} \cdot \overline{\mathsf{In}_\alpha(v,2)} \big) \Big) \\ & - m \cdot \overline{\mathsf{In}_\alpha(v,1)} \cdot \big(\overline{\mathsf{In}_\alpha(v,2)} - 1 + \mathsf{In}_\alpha(v,2) \big) \\ & + m \cdot \big(\mathsf{In}_\alpha(v,2) - 1 \big) \cdot \big(\overline{\mathsf{In}_\alpha(v,1)} - 1 + \mathsf{In}_\alpha(v,1) \big) \\ & + \big(1 - \mathsf{In}_\alpha(v,1) \big) \cdot q_2 \\ & + \big(1 - c_2 \big) \cdot q_1 \ , \end{aligned} \tag{40}$$

   where the first line uses Equation (11), and the following two lines uses negation axioms.

3. *v is an* and *gate ($m = m' \cdot \mathsf{IsAnd}(v)$, and $b_v = c_1 \cdot c_2$.* We have

$$\begin{aligned} p_v = & m' \cdot \mathsf{IsAnd}(v) \cdot \big(\mathsf{Out}_\alpha(v) - \mathsf{In}_\alpha(v,1) \cdot \mathsf{In}_\alpha(v,2) \big) \\ & + \mathsf{In}_\alpha(v,1) \cdot q_2 \\ & + c_2 \cdot q_1 \ , \end{aligned} \tag{41}$$

   where the first line uses Equation (12).

This completes the description of the SoS derivation of $\mathsf{Out}_\alpha(s) = b$. Observe that the final proof p_s is of degree $O(s)$: in every inductive step we increase the degree of the proof by at most a constant.

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